

7 (a) (capacitance =) charge/potential (difference) B1 [1]

(b) $V = V_1 + V_2 + V_3$ B1

either $Q/C = Q/C_1 + Q/C_2 + Q/C_3$ or $V/Q = V_1/Q + V_2/Q + V_3/Q$
and so $1/C = 1/C_1 + 1/C_2 + 1/C_3$ B1 [2]

(c) (i) 1. $1/C_T = (1/200) + (1/600)$
 $C_T = 150 \mu\text{F}$ A1 [1]

2. $Q = CV$
 $= 150 \times 10^{-6} \times 12$ or $600 \times 10^{-6} \times 3.0$ or $200 \times 10^{-6} \times 9.0$
 $= 1.8 \times 10^{-3} \text{ C}$ A1 [1]

3. $V = Q/C = 1.8 \times 10^{-3} / 600 \times 10^{-6}$ or $V = [200 / (200 + 600)] \times 12$
 $= 3(.0) \text{ V}$ A1 [1]

(ii) energy = $\frac{1}{2}CV^2$ or energy = $\frac{1}{2}QV$ and $C = Q/V$ C1
 $\frac{1}{2} \times C \times 3^2 = 2 \times \frac{1}{2} \times C \times V^2$ C1
 $V = 2.1 \text{ V}$ A1 [3]